

```
In [3]: #numpy
import numpy as np
```

```
In [2]: np.__version__
```

```
Out[2]: '2.3.0'
```

```
import sys sys.version
```

```
In [4]: np.arange(1,5)
```

```
Out[4]: array([1, 2, 3, 4])
```

```
In [6]: np.arange(20,10)
```

```
Out[6]: array([], dtype=int64)
```

```
In [7]: np.arange(-20,10)
```

```
Out[7]: array([-20, -19, -18, -17, -16, -15, -14, -13, -12, -11, -10, -9, -8,
              -7, -6, -5, -4, -3, -2, -1,  0,  1,  2,  3,  4,  5,
               6,  7,  8,  9])
```

```
In [8]: np.arange(10,10)
```

```
Out[8]: array([], dtype=int64)
```

```
In [9]: np.arange(10,3,5)
```

```
Out[9]: array([], dtype=int64)
```

```
In [10]: np.arange(19,30,5,8)
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[10], line 1
----> 1 np.arange(19,30,5,8)

TypeError: Cannot interpret '8' as a data type
```

```
In [11]: np.zeros(3)
```

```
Out[11]: array([0., 0., 0.])
```

```
In [12]: np.zeros(5,dtype=int)
```

```
Out[12]: array([0, 0, 0, 0, 0])
```

```
In [14]: np.zeros((2,2),dtype=int)
```

```
Out[14]: array([[0, 0],
               [0, 0]])
```

```
In [16]: zero=np.zeros([2,2])
print(zero)
```

```
[[0. 0.]  
 [0. 0.]]
```

```
In [19]: np.zeros(  
         (2,10))
```

```
Out[19]: array([[0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],  
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.]])
```

```
In [20]: n=(6,7)  
         n1=(6,8)  
         print(np.zeros(n1))
```

```
[[0. 0. 0. 0. 0. 0. 0. 0. 0.]  
 [0. 0. 0. 0. 0. 0. 0. 0. 0.]  
 [0. 0. 0. 0. 0. 0. 0. 0. 0.]  
 [0. 0. 0. 0. 0. 0. 0. 0. 0.]  
 [0. 0. 0. 0. 0. 0. 0. 0. 0.]  
 [0. 0. 0. 0. 0. 0. 0. 0. 0.]
```

```
In [21]: print(np.zeros(n,dtype=int))
```

```
[[0 0 0 0 0 0 0]  
 [0 0 0 0 0 0 0]  
 [0 0 0 0 0 0 0]  
 [0 0 0 0 0 0 0]  
 [0 0 0 0 0 0 0]  
 [0 0 0 0 0 0 0]]
```

```
In [22]: np.ones(2)
```

```
Out[22]: array([1., 1.])
```

```
In [23]: range(5)
```

```
Out[23]: range(0, 5)
```

```
In [26]: r=range(5)  
         r
```

```
Out[26]: range(0, 5)
```

```
for i in r: print(i)
```

```
In [30]: r=list(range(1,10,3))  
         r
```

```
Out[30]: [1, 4, 7]
```

```
In [29]: for i in r:  
         print(i)
```

```
1  
4  
7
```

```
In [31]: np.random.rand(5)
```

```
Out[31]: array([0.84604869, 0.79663631, 0.80084408, 0.95971861, 0.6590462 ])
```

```
In [32]: np.rand(4)
```

```
-----
AttributeError                                Traceback (most recent call last)
Cell In[32], line 1
----> 1 np.rand(4)

File ~\AppData\Roaming\Python\Python313\site-packages\numpy\__init__.py:808, in _
_getattr__(attr)
    805     import numpy.char as char
    806     return char.chararray
--> 808 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'rand'
```

```
In [33]: np.random.rand(2,4)
```

```
Out[33]: array([[0.67711716, 0.9758414 , 0.06396369, 0.38137791],
               [0.5828281 , 0.11225661, 0.39384584, 0.00205989]])
```

```
In [35]: np.random.randint(2,4)
```

```
Out[35]: 3
```

```
In [36]: np.random.randint(10,20,5)
```

```
Out[36]: array([10, 19, 17, 17, 12], dtype=int32)
```

```
In [38]: np.random.randint(0)
```

```
-----
ValueError                                Traceback (most recent call last)
Cell In[38], line 1
----> 1 np.random.randint(0)

File numpy/random/mtrand.pyx:794, in numpy.random.mtrand.RandomState.randint()

File numpy/random/_bounded_integers.pyx:2885, in numpy.random._bounded_integers._
rand_int32()

ValueError: high <= 0
```

```
In [39]: np.random.randint(-30,20,10)
```

```
Out[39]: array([ 16,   1, -12, -15, -20,   4,  -5, -22,  10, -19], dtype=int32)
```

```
In [40]: np.random.rand(2,10)
```

```
Out[40]: array([[0.45012112, 0.98145791, 0.68710874, 0.52141215, 0.13549447,
                0.1167739 , 0.1320463 , 0.79084104, 0.616552 , 0.87699338],
               [0.8808307 , 0.86632808, 0.24186034, 0.569824 , 0.71704303,
                0.73523501, 0.52801425, 0.13649096, 0.32392108, 0.99579451]])
```

```
In [41]: np.random.randint(10,40,(10,10))
```

```
Out[41]: array([[27, 26, 27, 24, 12, 17, 33, 36, 15, 32],
               [36, 13, 30, 32, 39, 32, 16, 39, 35, 25],
               [18, 29, 37, 13, 25, 14, 38, 23, 27, 34],
               [31, 21, 16, 16, 39, 28, 24, 20, 30, 12],
               [39, 22, 25, 28, 23, 28, 10, 33, 21, 37],
               [15, 28, 12, 38, 32, 17, 21, 33, 33, 19],
               [30, 29, 27, 12, 21, 28, 21, 36, 14, 13],
               [28, 23, 19, 32, 31, 38, 34, 17, 22, 38],
               [24, 26, 31, 37, 19, 34, 30, 19, 37, 10],
               [10, 11, 34, 17, 25, 14, 28, 25, 29, 25]], dtype=int32)
```

```
In [42]: np.arange(1,13).reshape(3,4)
```

```
Out[42]: array([[ 1,  2,  3,  4],
               [ 5,  6,  7,  8],
               [ 9, 10, 11, 12]])
```

```
In [44]: np.arange(1,13).reshape(12,1)
```

```
Out[44]: array([[ 1],
               [ 2],
               [ 3],
               [ 4],
               [ 5],
               [ 6],
               [ 7],
               [ 8],
               [ 9],
               [10],
               [11],
               [12]])
```

```
In [46]: b=np.random.randint(10,20,(5,4))
b
```

```
Out[46]: array([[16, 18, 15, 16],
               [15, 16, 19, 13],
               [19, 18, 11, 17],
               [19, 13, 19, 13],
               [11, 16, 12, 17]], dtype=int32)
```

```
In [47]: b[:]
```

```
Out[47]: array([[16, 18, 15, 16],
               [15, 16, 19, 13],
               [19, 18, 11, 17],
               [19, 13, 19, 13],
               [11, 16, 12, 17]], dtype=int32)
```

```
In [48]: b[1:3]
```

```
Out[48]: array([[15, 16, 19, 13],
               [19, 18, 11, 17]], dtype=int32)
```

```
In [49]: b[1,2]
```

```
Out[49]: np.int32(19)
```

```
In [51]: b[1,3]
```

```
Out[51]: np.int32(13)
```

```
In [52]: b[0:-2]
```

```
Out[52]: array([[16, 18, 15, 16],
                [15, 16, 19, 13],
                [19, 18, 11, 17]], dtype=int32)
```

```
In [56]: b[:3,:4]
```

```
Out[56]: array([[16, 18, 15, 16],
                [15, 16, 19, 13],
                [19, 18, 11, 17]], dtype=int32)
```

```
In [57]: b[0,2]
```

```
Out[57]: np.int32(15)
```

```
In [59]: b[-5,-3]
```

```
Out[59]: np.int32(18)
```

```
In [58]: b
```

```
Out[58]: array([[16, 18, 15, 16],
                [15, 16, 19, 13],
                [19, 18, 11, 17],
                [19, 13, 19, 13],
                [11, 16, 12, 17]], dtype=int32)
```

```
In [60]: b[-4,2]
```

```
Out[60]: np.int32(19)
```

```
In [61]: b
```

```
Out[61]: array([[16, 18, 15, 16],
                [15, 16, 19, 13],
                [19, 18, 11, 17],
                [19, 13, 19, 13],
                [11, 16, 12, 17]], dtype=int32)
```

```
In [63]: b[-4:2]
```

```
Out[63]: array([[15, 16, 19, 13]], dtype=int32)
```

```
In [62]: b
```

```
Out[62]: array([[16, 18, 15, 16],
                [15, 16, 19, 13],
                [19, 18, 11, 17],
                [19, 13, 19, 13],
                [11, 16, 12, 17]], dtype=int32)
```

```
In [65]: #operations
a=np.random.randint(10,20,10)
a
```

```
Out[65]: array([15, 16, 16, 18, 15, 11, 15, 12, 18, 10], dtype=int32)
```

```
In [4]: arr2=np.random.randint(0,100,(10,10))
arr2
```

```
Out[4]: array([[92, 85, 16, 32,  7, 47, 98, 72, 22, 40],
               [41, 79, 55, 64, 59, 74, 52, 37, 81, 79],
               [48, 61, 55, 62, 67, 37, 82, 99, 54, 12],
               [39, 40, 64, 52, 19, 78, 15, 81, 33, 78],
               [66, 48, 75, 18, 29, 52, 84, 43, 59, 72],
               [62, 75, 98, 55, 82, 40, 16, 30, 45,  5],
               [ 4, 21, 34, 16, 96, 46,  9, 93, 32, 49],
               [91,  3, 95, 69, 56, 37, 11, 49, 73, 45],
               [73, 94, 13, 82,  5, 76,  8, 72, 86,  0],
               [90, 78, 60,  7, 57, 55, 41, 39, 83,  8]], dtype=int32)
```

```
In [76]: arr2[:,]
```

```
Out[76]: array([[82, 96, 14, 64, 63, 48, 58, 20, 28, 18],
               [11, 88, 76, 36, 96, 72, 44, 10, 59, 46],
               [23, 82,  7, 16, 56, 21, 13, 20, 81, 56],
               [77, 37, 88, 96, 51, 63, 77, 86, 39, 52],
               [10, 12, 19, 59,  9, 52, 11, 87,  5, 19],
               [77, 76,  5, 37, 27, 86, 74,  1, 90, 34],
               [49, 33, 49, 57, 18, 70, 72, 96, 36, 46],
               [41, 96, 43, 31, 13, 19,  3, 79, 21,  5],
               [65,  2, 44, 54, 39, 52, 80, 20, 46, 91],
               [68, 51, 14, 73, 78, 85, 94, 60,  7, 13]], dtype=int32)
```

```
In [78]: arr2[:4]
```

```
Out[78]: array([[82, 96, 14, 64, 63, 48, 58, 20, 28, 18],
               [11, 88, 76, 36, 96, 72, 44, 10, 59, 46],
               [23, 82,  7, 16, 56, 21, 13, 20, 81, 56],
               [77, 37, 88, 96, 51, 63, 77, 86, 39, 52]], dtype=int32)
```

```
In [80]: arr2[1,4]
```

```
Out[80]: np.int32(96)
```

```
In [81]: arr2[-5,-5]
```

```
Out[81]: np.int32(86)
```

```
In [7]: arr2[:,]
```

```
Out[7]: array([[92, 85, 16, 32,  7, 47, 98, 72, 22, 40],
               [41, 79, 55, 64, 59, 74, 52, 37, 81, 79],
               [48, 61, 55, 62, 67, 37, 82, 99, 54, 12],
               [39, 40, 64, 52, 19, 78, 15, 81, 33, 78],
               [66, 48, 75, 18, 29, 52, 84, 43, 59, 72],
               [62, 75, 98, 55, 82, 40, 16, 30, 45,  5],
               [ 4, 21, 34, 16, 96, 46,  9, 93, 32, 49],
               [91,  3, 95, 69, 56, 37, 11, 49, 73, 45],
               [73, 94, 13, 82,  5, 76,  8, 72, 86,  0],
               [90, 78, 60,  7, 57, 55, 41, 39, 83,  8]], dtype=int32)
```

```
In [6]: arr2[:,-3]
```

```
Out[6]: array([[92, 85, 16, 32,  7, 47, 98, 72, 22, 40],
               [41, 79, 55, 64, 59, 74, 52, 37, 81, 79],
               [48, 61, 55, 62, 67, 37, 82, 99, 54, 12],
               [39, 40, 64, 52, 19, 78, 15, 81, 33, 78],
               [66, 48, 75, 18, 29, 52, 84, 43, 59, 72],
               [62, 75, 98, 55, 82, 40, 16, 30, 45,  5],
               [ 4, 21, 34, 16, 96, 46,  9, 93, 32, 49]], dtype=int32)
```

```
In [8]: arr2.max()
```

```
Out[8]: np.int32(99)
```

```
In [9]: from numpy import *
a=array([1,2,3,4,5,6])
median(a)
```

```
Out[9]: np.float64(3.5)
```

```
In [12]: arr=np.array([0,1,2,3,4,5])
arr
type(arr)
```

```
Out[12]: numpy.ndarray
```

```
In [13]: arr.reshape(3,2)
```

```
Out[13]: array([[0, 1],
               [2, 3],
               [4, 5]])
```

```
In [14]: arr.reshape(2,3)
```

```
Out[14]: array([[0, 1, 2],
               [3, 4, 5]])
```

```
In [18]: arr.reshape(2,3,order='A')
```

```
Out[18]: array([[0, 1, 2],
               [3, 4, 5]])
```

```
In [19]: arr.reshape(3,3)
```

```
-----
ValueError                                Traceback (most recent call last)
Cell In[19], line 1
----> 1 arr.reshape(3,3)

ValueError: cannot reshape array of size 6 into shape (3,3)
```

```
In [21]: #indexing
mat=np.arange(0,100).reshape(10,10)
mat
```

```
Out[21]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
               [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
               [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
               [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
               [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
               [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
               [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
               [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
               [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
               [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [22]: mat[4,5]
```

```
Out[22]: np.int64(45)
```

```
In [23]: mat[:,6]
```

```
Out[23]: array([ 6, 16, 26, 36, 46, 56, 66, 76, 86, 96])
```

```
In [24]: mat[:,8]
```

```
Out[24]: array([ 8, 18, 28, 38, 48, 58, 68, 78, 88, 98])
```

```
In [25]: mat[:, -1]
```

```
Out[25]: array([ 9, 19, 29, 39, 49, 59, 69, 79, 89, 99])
```

```
In [26]: mat[1:4]
```

```
Out[26]: array([[10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
               [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
               [30, 31, 32, 33, 34, 35, 36, 37, 38, 39]])
```

```
In [27]: mat[1,4]
```

```
Out[27]: np.int64(14)
```

```
In [28]: mat[3:-3]
```

```
Out[28]: array([[30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
               [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
               [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
               [60, 61, 62, 63, 64, 65, 66, 67, 68, 69]])
```

```
In [29]: mat[0:10:3]
```

```
Out[29]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
               [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
               [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
               [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [30]: mat[:, -3]
```

```
Out[30]: array([[90, 91, 92, 93, 94, 95, 96, 97, 98, 99],
               [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
               [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
               [ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9]])
```



```
In [31]: mat
```

```
Out[31]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
               [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
               [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
               [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
               [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
               [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
               [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
               [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
               [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
               [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])

mat[2:6,2:4]
```

```
In [32]: mat[2:6,2:4]
```

```
Out[32]: array([[22, 23],
               [32, 33],
               [42, 43],
               [52, 53]])
```

```
In [33]: mat[0,1]
```

```
Out[33]: np.int64(1)
```

```
In [34]: #masking
mat
```

```
Out[34]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
               [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
               [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
               [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
               [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
               [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
               [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
               [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
               [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
               [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [36]: id(mat)
```

```
Out[36]: 2147280014768
```

```
In [37]: mat[mat<50]
```

```
Out[37]: array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
               17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33,
               34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49])
```

```
In [38]: mat
```

```
Out[38]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
                [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
                [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
                [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
                [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
                [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
                [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
                [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
                [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [39]: mat>50
```

```
Out[39]: array([[False, False, False, False, False, False, False, False, False,
                False],
                [False, False, False, False, False, False, False, False, False,
                False],
                [False, False, False, False, False, False, False, False, False,
                False],
                [False, False, False, False, False, False, False, False, False,
                False],
                [False, False, False, False, False, False, False, False, False,
                False],
                [False, True,  True,  True,  True,  True,  True,  True,  True,
                True],
                [ True,  True,  True,  True,  True,  True,  True,  True,  True,
                True],
                [ True,  True,  True,  True,  True,  True,  True,  True,  True,
                True],
                [ True,  True,  True,  True,  True,  True,  True,  True,  True,
                True],
                [ True,  True,  True,  True,  True,  True,  True,  True,  True,
                True]])
```

```
In [40]: mat[mat==50]
```

```
Out[40]: array([50])
```

```
In [42]: a1=mat[mat<50]
a1
```

```
Out[42]: array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
                17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33,
                34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49])
```

```
In [ ]: #pythonm programme to generate otp
```

```
In [48]: import random
def generate_otp(length=4):
    digits='012345'
    otp=''.join(random.choice(digits) for _ in range(length))
    return otp
otp_length=4
otp=generate_otp(otp_length)
print(f"your otp is :{otp}")
```

your otp is :2115